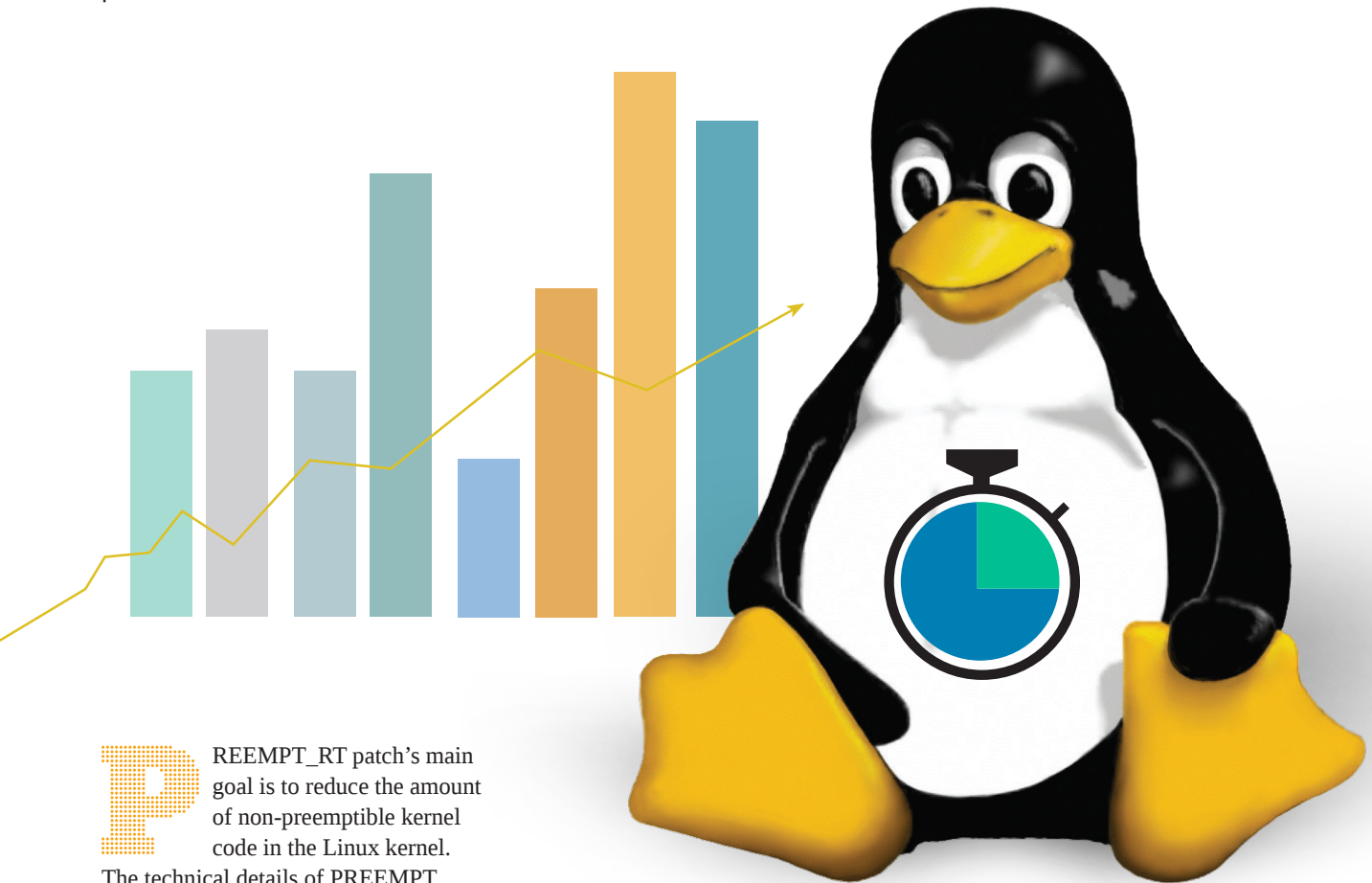


An Analysis of the Real-Time Performance of Linux Kernels

The main focus of this work is to measure and analyse the real-time performance of the vanilla Linux kernel as against the *PREEMPT_RT* patched Linux kernel, using the rt-tests test suite. The results show that the *PREEMPT_RT* patch kernel has a better performance than the vanilla kernel.



PREEMPT_RT patch's main goal is to reduce the amount of non-preemptible kernel code in the Linux kernel.

The technical details of PREEMPT_RT patch are available at https://wiki.linuxfoundation.org/realtime/documentation/technical_details/start.

RT-patched Linux kernel installation

The kernel patch can be downloaded from <https://mirrors.edge.kernel.org/pub/linux/kernel/projects/rt/> and the vanilla Linux kernel can be downloaded

from <https://mirrors.edge.kernel.org/pub/linux/kernel/>. It is important to make sure that the version of rt-patch and the kernel are the same.

Now extract the kernel sources and apply the rt-patch. In the *config* options, set the 'Fully Preemptible kernel (RT)' option. Next, build the

kernel, update the grub boot loader and reboot into the newly installed real-time patched kernel.

When the command *uname -a* is run after booting the rt-patched kernel, the output should contain *SMP PREEMPT RT*, to validate that the system is running on a real-time kernel.